

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 647 Image Synthesis — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Physically Based Rendering, 4th ed. — free online	The primary text, and the most complete free resource in graphics. It is a literate program: the book is the renderer. Read the chapter before each module and read the code with it.
Ray Tracing in One Weekend series (free)	Three short books that get a working path tracer on screen fast. Do the first one before Module 03 — getting something rendering early makes the theory concrete.
TU Wien — Rendering / Light Transport (free video)	Károly Zsolnai-Fehér's full graduate course. Excellent on the intuition behind Monte Carlo and on the comparative strengths of bidirectional methods.
Eric Veach — Robust Monte Carlo Methods for Light Transport (thesis, free)	Where multiple importance sampling, BDPT, and Metropolis light transport come from. Chapter 9 on MIS is the clearest exposition that exists and is worth reading in the original.
Scratchapixel — rendering lessons (free)	Slower and more elementary than PBRT, with the algebra written out. Useful when a PBRT derivation moves too fast.
Benedikt Bitterli — rendering resources and test scenes (free)	Scenes in PBRT format with reference renders. Your validation set from Module 08 onward.

Module-by-module map

Module 01 — Radiometry: What Rendering Computes

Resource	What it covers for this module
PBRT 4th ed. — Chapter 4, Radiometry, Spectra, and Colour	The definitive treatment of this module. §4.1 for the quantities and §4.2 for radiance invariance. Skip the colour sections until Module 13.
Scratchapixel — Mathematics of Shading / solid angle	Slower than PBRT with the algebra written out. The best free explanation of solid angle if the PBRT version moves too fast.
TU Wien Rendering — early lectures on radiometry	The intuition for why radiance is the transported quantity, developed visually.
PBRT 4th ed. — §5.4, The Camera Measurement Equation	The pixel integral of §5, derived properly, including the importance function that makes bidirectional methods possible.

Module 02 — The Rendering Equation

Resource	What it covers for this module
Kajiya — The Rendering Equation (1986, free PDF)	The original paper. Short, readable, and worth reading in the original — the framing of the problem is the contribution.
PBRT 4th ed. — Chapter 13, Light Transport I: Surface Reflection	The rendering equation, both forms, the path-integral formulation, and the derivation of the geometry term.
Veach thesis — Chapter 8, The Path Integral Formulation	The cleanest statement of the measurement equation and the path integral. Read it before Module 11; skim it now.
TU Wien Rendering — the rendering equation lectures	The Neumann-series-as-path-lengths interpretation, built up visually. Good complement to §2.

Module 03 — Rays and Intersection

Resource	What it covers for this module
PBRT 4th ed. — Chapter 6, Shapes	Intersection for every primitive type, and §6.8 on managing rounding error — the most careful free treatment of §1 and §3 anywhere.
Woop, Benthin & Wald — Watertight Ray/Triangle Intersection (free)	The source for §2. Short paper, complete algorithm, and the failure analysis is worth reading on its own.
Ray Tracing Gems — Chapter 6, A Fast and Robust Method for Avoiding Self-Intersection (free)	The ULP-offset method of §3, with the full code. The whole book is free.
Igehy — Tracing Ray Differentials (1999, free)	The original derivation of §4, including propagation through reflection and refraction.
Ray Tracing in One Weekend	Work through this now if you have not. It gets an image on screen in a few hours and makes the rest of the course concrete.

Module 04 — Acceleration Structures

Resource	What it covers for this module
PBRT 4th ed. — Chapter 7, Primitives and Intersection Acceleration	BVH construction including binned SAH, with complete code. The primary reading for this module.
Wald, Boulos & Shirley — Ray Tracing Deformable Scenes with Dynamic BVHs (free)	Where much of the modern BVH consensus comes from, including refitting and the case against kd-trees.
Stich, Friedrich & Dietrich — Spatial Splits in BVHs (SBVH, free)	The spatial-split hybrid of §5. Short and clearly argued.
Embree documentation and source	A production BVH you can read. Compare your traversal loop against theirs after you have written one.
Ray Tracing Gems — chapters on BVH traversal and compression (free)	Practical implementation detail, especially for the GPU case.

Module 05 — Monte Carlo Integration

Resource	What it covers for this module
PBRT 4th ed. — Chapter 2, Monte Carlo Integration	The estimator, variance, and the sampling machinery, with proofs. The primary reading.
Veach thesis — Chapter 2, Monte Carlo Integration	More careful than most treatments on what unbiasedness does and does not guarantee.
TU Wien Rendering — Monte Carlo lectures	The intuition for why dividing by the PDF works, built up slowly. Worth watching before reading PBRT.
PBRT — the χ^2 test in the test suite	A working implementation of the sampler validation in §4. Copy the approach.

Module 06 — Sampling and Variance Reduction

Resource	What it covers for this module
PBRT 4th ed. — Chapter 2.2–2.4 and Chapter 8, Sampling and Reconstruction	Inversion, the standard sampling routines, stratification, and the Sobol sampler with complete code.
PBRT — A.5, Sampling Algorithms	The reference list of how to sample every common distribution in rendering. Worth bookmarking.
Veach thesis — Chapter 2 on variance reduction	Careful treatment of importance sampling, including when it fails and why defensive sampling is sometimes necessary.
Christensen & Jarosz — The Path to Path-Traced Movies (free)	What production renderers actually do about sampling, stratification, and roulette. Useful reality check on §5.

Resource	What it covers for this module
Heitz — blue-noise sampling papers (free)	The perceptual argument of §3 developed properly, with sample sets you can use.

Module 07 — BSDFs and Microfacet Theory

Resource	What it covers for this module
PBRT 4th ed. — Chapters 9 and 14, Reflection Models and Sampling	The complete treatment with code: Fresnel, microfacet distributions, Smith masking, and the sampling routines.
Walter et al. — Microfacet Models for Refraction through Rough Surfaces (2007, free)	The paper that brought GGX into graphics, including the rough transmission derivation of §3.
Heitz — Sampling the GGX Distribution of Visible Normals (2018, free)	The algorithm of §2, with the derivation and complete code. Short paper, large practical payoff.
Heitz — Understanding the Masking-Shadowing Function (2014, free)	Everything about G, including why the height-correlated form is the right default.
Kulla & Conty — Revisiting Physically Based Shading at Imageworks (free)	The energy compensation of §1, from the people who deployed it. The whole SIGGRAPH shading course archive is free and excellent.

Module 08 — Path Tracing and Multiple Importance Sampling

Resource	What it covers for this module
PBRT 4th ed. — Chapter 13, Light Transport I	Path tracing, NEE, and MIS with complete code. §13.10 on MIS is the implementation to compare yours against.
Veach & Guibas — Optimally Combining Sampling Techniques (1995, free)	The original MIS paper. Short, and the figure comparing strategies across light sizes is the one to reproduce.
Veach thesis — Chapter 9	The fuller treatment, including the optimality proof for the balance heuristic.
Benedikt Bitterli — rendering resources	Scenes in PBRT format with reference images. Your validation set for §4.

Module 09 — Lights and Sampling Them

Resource	What it covers for this module
PBRT 4th ed. — Chapter 12, Light Sources	Every light type with its sampling routine and PDF, including the environment map 2D distribution of §3.
PBRT 4th ed. — §12.6, Light Sampling (BVH)	The light BVH of §4, with the importance heuristic and the orientation cone bounds.

Resource	What it covers for this module
Ureña, Fajardo & King — An Area-Preserving Parametrization for Spherical Rectangles (free)	Solid-angle sampling for quads. Arnold's research page has a great deal of production-grade free material.
Bitterli et al. — Spatiotemporal Reservoir Resampling (ReSTIR, free)	The many-lights result of §4. Read it after Module 12.
Poly Haven — free HDRIs (CC0)	High-quality environment maps, CC0 licensed, for testing §3.

Module 10 — Participating Media

Resource	What it covers for this module
PBRT 4th ed. — Chapters 11 and 14, Volume Scattering and Light Transport II	The coefficients, phase functions, and the complete delta-tracking and ratio-tracking implementations.
Novák et al. — Monte Carlo Methods for Volumetric Light Transport (survey, free)	The definitive survey. Everything in §3, with the unbiasedness proofs and the full tracking family.
Fong et al. — Production Volume Rendering (SIGGRAPH course, free)	How volumes are actually rendered in film, including majorant grids and the practical compromises.
Wrenninge — Production Volume Rendering (book site)	Course notes and code, free. Strong on the data structures.
Disney — Cloud data set (CC0)	A production cloud volume, free to use. Your test asset for Project 2.

Module 11 — Bidirectional Methods

Resource	What it covers for this module
Veach thesis — Chapters 10 and 11, BDPT and MLT	The original source for both methods, including the measurement equation and the camera importance function of §2.
PBRT 4th ed. — Chapter 13 and the online BDPT chapter	BDPT with complete code, including the MIS weight computation, which is the part that is hard to get right.
Jensen — Realistic Image Synthesis Using Photon Mapping	Photon mapping from its author; the papers are free on this page.
Hachisuka & Jensen — Progressive Photon Mapping (free)	The method that removed photon mapping's memory limit while keeping consistency.
Kelemen et al. — Primary Sample Space MLT (free)	The far simpler MLT variant mentioned in §5. Start here if you implement one.

Module 12 — Denoising and Adaptive Sampling

Resource	What it covers for this module
Schied et al. — Spatiotemporal Variance-Guided Filtering (SVGF, free)	The filter of §2, including the variance estimation and temporal accumulation. The paper to implement from.
Intel Open Image Denoise	A free, production-quality learned denoiser with a simple API. Use it as the comparison point for your own filter.
Bitterli et al. — Spatiotemporal Reservoir Resampling (ReSTIR, free)	The §4 paper. Read the supplemental material for the bias discussion.
Wyman & Panteleev › A Gentle Introduction to ReSTIR (SIGGRAPH course, free)	Far more approachable than the paper. Start here.
Zwicker et al. — Recent Advances in Adaptive Sampling and Reconstruction (survey, free)	The survey covering §1 and §2, including the bias analysis of adaptive sampling.

Module 13 — Spectral Rendering, Cameras, and Getting the Image Right

Resource	What it covers for this module
PBRT 4th ed. — Chapters 4.5 and 5, Colour and Cameras	Spectral representation, the CIE observer, colour spaces, the thin lens model, and the film and filter machinery.
Wilkie et al. — Hero Wavelength Spectral Sampling (2014, free)	The technique of §1, with the MIS derivation.
Cornell Box data and measurements	Geometry, measured spectral reflectances, and radiometric measurements. The physical validation of §4.
Troy Sobotka — AgX and colour management writing	The display transform of §3, with a clear account of why hue preservation in highlights matters.
Poynton — Frequently Asked Questions about Gamma (free)	The clearest explanation of the transfer function, and why calling it 'gamma' causes so much confusion.